PROJECT TITLE

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

Course Code & Course Name

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING (BRANCH)

Submitted by

Student 1 Name (Registration Number)

Student 2 Name (Registration Number)

Under the Supervision of

Guide Name

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Project Title" has been carried out by Student 1 Name (Reg. No) and Student 2 Name (Reg. No) under the supervision of Guide Name and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR	COURSE FACULTY
Title / Name, Designation Branch / Department Name SIMATS Engineering, SIMATS, Chennai – 602105.	Title / Name, Designation Branch / Department Name SIMATS Engineering, SIMATS, Chennai – 602105.

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, [Student Names] of the [Department Name], SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "[Project Title]" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Students with Names

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Student 1	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Student 2	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Student 3	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Student 4	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Total					100%

ABSTRACT

Maximum approximately **300–400 words**.

The abstract shall contain:

- Engineering problem
- Need/significance
- Proposed solution
- Methodology
- Major results
- Principal conclusion

Keywords

4–6 keywords.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Provide the technical and application context of the project.

1.2 Need for the Project

Explain:

- Why the problem needs to be addressed
- Who is affected by the problem
- Existing limitations
- Engineering significance

1.3 Problem Statement

Provide a precise engineering problem statement.

The problem should, wherever appropriate, involve:

- Multiple interacting factors
- Technical complexity
- Conflicting requirements
- Realistic engineering constraints

1.4 Project Aim

State the overall aim.

1.5 Project Objectives

Provide measurable objectives.

Example:

1. To design...
2. To develop...
3. To model...
4. To implement...
5. To test...
6. To evaluate...

1.6 Scope

Clearly state:

- What is included
- What is excluded
- Assumptions
- Boundary conditions

1.7 Expected Engineering Outcome

Define the intended:

- System
- Product
- Process
- Prototype
- Algorithm
- Model
- Device
- Infrastructure solution

as applicable.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Briefly describe how relevant literature, standards, patents, products or existing technologies were identified.

2.2 Review of Existing Work

Critically review relevant:

- Research papers
- Existing systems
- Commercial solutions
- Technologies
- Methods
- Patents, where applicable

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work

2.4 Research / Engineering Gap

Clearly identify what remains unresolved.

2.5 Proposed Contribution

Explain how the project differs from or improves upon existing approaches.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target

3.2 Constraints & Applicable Standards

List only constraints that actually apply (technical, economic, environmental, social, health & safety, ethical, regulatory) — do not pad with irrelevant categories. Name the specific standard, not just 'IEEE standards were followed.'

Standard / Code	Requirement	How Applied in This Project

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority

3.4 Alternative Solutions & Evaluation

Describe each alternative in 2–3 sentences max, then let the evaluation matrix carry the comparison — do not repeat the same comparison in prose afterward.

Alternative 1: [2–3 sentence description]

Alternative 2: [2–3 sentence description]

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance				
Cost				
Safety				
Sustainability				
Reliability				

3.5 Selected Design & Detailed Design

Justify the selection with evidence from 3.4, not preference. Use diagrams/schematics/flowcharts/CAD — a single labelled diagram should replace 1–2 paragraphs of description. Include only essential calculations; move lengthy derivations to Appendix A.

[Justification for final design selection.]

[Insert architecture diagram / schematic / flowchart here.]

[Key engineering calculations — essential ones only.]

[Systems approach: how subsystems interface and depend on each other — 3–4 sentences.]

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification

3.7 Sustainability, Ethics, Safety, Risk & Security

This is the section institutional guidelines flag most for vague statements. Never write 'sustainability was considered' — the table must show what evidence was evaluated and how it changed the design decision.

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability			
Ethics			
Health & Safety			
Security			

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

One paragraph on methodology, one line/table on materials, components, dataset, or resources used.

[Development approach in 3–4 sentences.]

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Use subheadings (not chapters) for whichever of hardware/software/fabrication apply. Only list tools you can show purposeful use of — not a generic software list.

[Implementation details — hardware/software/fabrication as applicable.]

Tool / Software / Equipment	Purpose	Stage Used

[Insert prototype photo / screenshot / final implementation evidence here.]

4.3 Test Plan & Verification

Define the test plan BEFORE presenting results.

<table>
<tr><th colspan="2">Requirement</th><th colspan="2">Test Method</th><th colspan="2">Acceptance Criterion</th></tr>
<tr><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr>
<tr><th>Specification</th><th colspan="2">Required Value</th><th colspan="2">Achieved Value</th><th>Status (Pass/Fail)</th></tr>
<tr><td></td><td colspan="2"></td><td colspan="2"></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

4.4 Results

Present measurements, graphs, and key data. Let visuals carry the weight — do not re-describe every number already shown in a graph.

[Insert results tables / graphs / simulation outputs here.]

4.5 Data Analysis & Engineering Judgment

Interpret the data (statistical/error/accuracy/performance metrics as relevant). Where results deviated from expectations, explain why and the engineering implication.

[Analysis and judgment discussion here.]

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)

4.7 Strengths & Limitations

State limitations honestly — do not claim the system is perfect.

Strengths: [bullets]

Limitations: [bullets]

4.8 Project Planning, Roles & Budget

Gantt chart or milestone table, roles/contribution table, and a brief cost table — this replaces the old standalone Project Management chapter.

[Insert Gantt chart / milestone timeline here.]

Student	Assigned Responsibility	Actual Contribution
Item	**Estimated Cost**	**Actual Cost**

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

[Conclusion here.] - ***Summarise the problem, solution, achievements, validation, and overall contribution — no new information here.***

5.2 Future Work

[Future work here.]

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired: [bullets]

Engineering & Professional Skills Developed: [bullets]

Individual Reflection (each student, 4–5 lines):

- **Most difficult engineering problem encountered and how it was solved**
- **A key engineering decision made personally and the evidence behind it**
- **New knowledge acquired independently**
- **What would be redesigned if repeated**

REFERENCES

Use a consistent recognised referencing style such as IEEE.

All:

- Literature
- Standards
- Websites
- Datasets
- Software
- Open-source resources
- AI-assisted resources, where applicable

shall be appropriately acknowledged.

APPENDICES

Appendix A – Detailed Calculations

Appendix B – Engineering Drawings / Schematics

Appendix C – Source Code / Code Repository Information

Only essential code excerpts should normally be included in the report. Complete source code may be maintained separately in the approved repository.

Appendix D – Datasheets

Appendix E – Experimental / Raw Data

Appendix F – Ethics / Institutional Approval

Where applicable.

Appendix G – Project Meeting / Progress Records

Appendix H – User/Industry Feedback

Where applicable.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Where applicable.

Appendix J – Individual Contribution Evidence

Mandatory for team projects.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)